WEEKLY BUSINESS REVIEW

Brazilian E-Commerce Marketplace | Olist Public Dataset (2016–2018)

1. Pipeline overview

The Python ELT pipeline loads seven source CSV files into raw tables, creates staging tables with parsed timestamps and numeric fields, and builds **mart_order_summary** at one row per order. Item and payment amounts are aggregated by order before joining customer and order attributes. Target tables are replaced on each run to avoid appending duplicate rows; processing and validation outcomes are logged.

Loaded data: 99,441 orders; 112,650 order-item rows; 96,096 distinct non-null customer identifiers (`customer_unique_id`).

2. Data-quality results

Check	Failures	Treatment / interpretation
Null order IDs	0	No failures observed in this run.
Duplicate order ID groups	0	No failures observed in this run.
Delivery before purchase	0	No failures observed in this run.
Orders without matching customer	0	No failures observed in this run.
Payment/item + freight mismatches (> R$1)	249	Flag for investigation. Preserve original records; do not disable the check or assume the cause without further evidence.
Invalid review scores	0	No failures observed in this run.

Payment reconciliation note: the check compares summed payment value with summed item price plus freight and flags differences greater than R$1. It identified 249 order groups. These records are retained and flagged; the underlying cause has not yet been established. The pipeline currently raises a quality alarm for this condition, so the run should not be described as fully passing all quality checks.

3. Pipeline reliability

Idempotency: the pipeline replaces raw, staging and mart tables rather than appending rows. **Failure handling:** missing files or required columns raise errors; quality-check failures are logged and cause a non-zero exit. A successful table build and a successful quality validation are separate outcomes.

4. Metrics and analytical highlights

Repeat-customer rate: 3.12% (2,997 repeat customers out of 96,096 distinct customers). A repeat customer is defined as a customer_unique_id with at least two distinct orders.

Revenue definition: merchandise revenue is the sum of item price, excluding freight. The monthly revenue query excludes canceled and unavailable orders. Month-over-month growth is (current month revenue – previous month revenue) / previous month revenue × 100. Growth can be unusually large when the previous month has very low revenue.

Metric	Formula	Purpose
North Star: fulfilled order rate	Delivered orders ÷ all orders × 100	Tracks the share of orders successfully fulfilled.
Merchandise revenue	SUM(item price) for eligible orders	Measures merchandise sales, excluding freight.
On-time delivery rate	Delivered orders with actual date ≤ estimated date ÷ delivered orders with both dates × 100	Tracks delivery reliability.
Average review score	AVG(valid review_score)	Customer-experience guardrail.
Repeat-customer rate	Customers with ≥2 orders ÷ distinct customer_unique_id × 100	Tracks customer retention.

5. Charts

The charts below are generated from the local SQLite database. Review scores and delivery-delay groups are descriptive; small groups should not be overinterpreted.

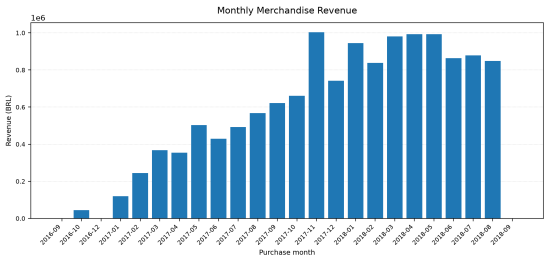


Figure 1. Monthly merchandise revenue

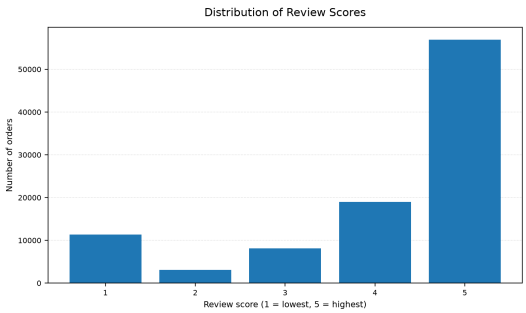


Figure 2. Review-score distribution

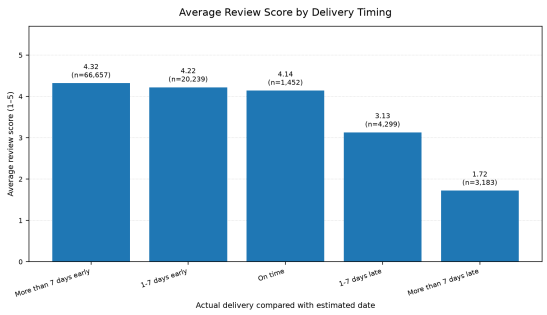


Figure 3. Review score by delivery delay

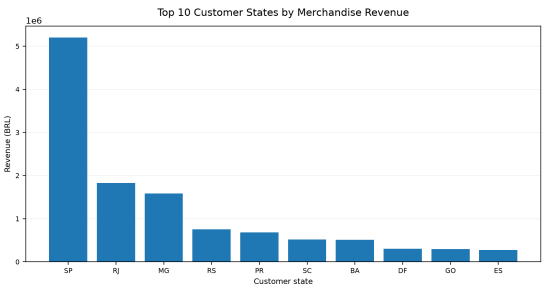


Figure 4. Top 10 states by merchandise revenue

6. Memo to the Vice President

Subject: A measured pilot to improve delivery reliability

The order-level analysis combines actual delivery timestamps, estimated delivery timestamps and review scores to examine whether delivery performance is associated with customer satisfaction. The operational review should compare the number of late orders with on-time/early orders and their average review scores, then identify seller or geographic segments with frequent delivery exceptions. These comparisons show association, not proof of causation.

Recommendation: run a four-week delivery-exception pilot. Prioritise orders predicted to miss their estimated delivery dates, review exception segments weekly, and track on-time delivery rate and average review score against a documented baseline. Expand the intervention only if the observed metrics improve without degrading other service measures.

Decision requested: approve the four-week pilot and weekly monitoring. Keep the payment-reconciliation alarm visible until the 249 mismatches have been investigated and the validation rule has been confirmed.

7. SQL analysis delivered

The project includes SQL queries for (1) monthly revenue and month-over-month growth, (2) top-three product categories by customer state, (3) a seven-calendar-day rolling average of daily orders, (4) repeat-customer rate using customer_unique_id, and (5) delivery delay relative to estimate versus average review score.

Limitations: the Olist dataset is observational and covers 2016–2018. Merchandise revenue is not profit. Delivery/review relationships can be affected by order mix, seller, geography and other factors. The payment discrepancies remain unresolved and should be presented as a known data-quality issue.